Coordinate Geometry Maze ANSWER KEY

Start top-left. Give distances in simplest radical form. Rotations and dilations are centered at the origin; k is the scale factor. Go through the door with your answer.

START

1 Find the midpoint.
(2, 4) and (8, 10)

(5, 7)

2 Find the distance.
(1, 2) and (4, 6)

$\sqrt{7}$

Find the midpoint.
(−3, 5) and (7, −1)

(4, 4)

Reflect over the *y*-axis.
(−6, 2)

(3, 3)

5

7

(5, −3)

(10, −6)

4

(−6, −2)

3 Reflect over the *x*-axis.
(3, −2)

(3, 2)

4 Translate the point.
(−4, 1)
$(x, y) \to (x + 5, y - 3)$

(−9, −2)

Dilate from the origin.
$(-2, 8)$, $k = \frac{1}{2}$

(−4, 16)

Find the slope.
perpendicular to
$y = -4x + 1$

(−3, −2)

(1, −2)

(1, 4)

(−9, 4)

$2\sqrt{5}$

(1, −4)

$-\frac{1}{4}$

5 Find the slope.
parallel to
$y = 3x - 4$

−3

8 Find the slope.
perpendicular to
$2x + 3y = 6$

$\frac{3}{2}$

9 Find the distance.
(−2, 3) and (4, −1)

$4\sqrt{13}$

Rotate about the origin.
(3, 4)
180°

3

(1, −5)

(−5, 2)

10

$2\sqrt{13}$

(5.5, −4.5)

(−4, 3)

6 Reflect over $y = x$.
(5, −1)

(−1, 5)

7 Rotate about the origin.
(2, 5)
90° counterclockwise

(5, −2)

10 Dilate from the origin.
(4, −6), $k = \frac{3}{2}$

(6, −9)

FINISH

You made it!

Coordinate Geometry Maze: Answer Grid **ANSWER KEY**

Write each answer in the order you found it, and show your work for that room.

1	Answer: **(5, 7)**	$(\frac{2+8}{2}, \frac{4+10}{2}) \rightarrow (5, 7)$
2	Answer: **5**	$d = \sqrt{3^2 + 4^2} \rightarrow d = \sqrt{25} = 5$
3	Answer: **(3, 2)**	$(x, y) \rightarrow (x, -y) \rightarrow (3, 2)$
4	Answer: **(1, −2)**	$(-4 + 5, 1 - 3) \rightarrow (1, -2)$
5	Answer: **3**	parallel: same slope $\rightarrow m = 3$
6	Answer: **(−1, 5)**	$(x, y) \rightarrow (y, x) \rightarrow (-1, 5)$
7	Answer: **(−5, 2)**	$(x, y) \rightarrow (-y, x) \rightarrow (-5, 2)$
8	Answer: $\frac{3}{2}$	$y = -\frac{2}{3}x + 2 \rightarrow$ perpendicular: $m = \frac{3}{2}$
9	Answer: $2\sqrt{13}$	$d = \sqrt{6^2 + 4^2} \rightarrow d = \sqrt{52} = 2\sqrt{13}$
10	Answer: **(6, −9)**	$(x, y) \rightarrow (\frac{3}{2}x, \frac{3}{2}y) \rightarrow (6, -9)$